

ADITI TRIPATHI

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leetcode.com/u/aditi3003

Career Objective

B.Tech Computer Science and Engineering (Artificial Intelligence) undergraduate with expertise in Machine Learning, Deep Learning, and Data Structures Algorithms. Passionate about designing intelligent systems, building scalable ML models, and leveraging data to solve real-world problems.

Education

Pranveer Singh Institute of Technology, Kanpur **Oct 2023 - June 2027**
Bachelor of Technology in Computer Science and Engineering [Percentage: 81](Till 6th Semester) Kanpur, India

Deoria Senior Secondary School **Apr 2021 - Mar 2022**
Senior Secondary (Class XII), CBSE Board [Percentage: 92.4] Deoria, Uttar Pradesh, India

Central Academy **Apr 2019 - Mar 2020**
Secondary (Class X), CBSE Board [Percentage: 91.6] Deoria, Uttar Pradesh, India

Technical Skills

Languages: C, C++, JavaScript, HTML, CSS, Python (ML/Data Analysis)

Concepts: Data Structures & Algorithms, Object-Oriented Programming, SQL, Machine Learning, Deep Learning, Natural Language Processing

Software Engineering: Engineering practices, Debugging, Scalability, Performance Optimization

Frameworks and Libraries: Keras, TensorFlow, scikit-learn, Matplotlib, Pandas

Developer Tools: MySQL, Git, VS Code, Google Colab, Jupyter Notebook, GitHub

PROJECTS

CompNet – Intelligent Network Congestion Control System  **Jan 2026**

- Developed an intelligent congestion control system using Machine Learning/Deep Learning models to predict network congestion in real time with **95% accuracy**.
- Designed a data-driven pipeline including preprocessing, feature extraction, and model training on network performance metrics such as throughput, delay, and packet loss.
- Integrated the model with simulated network environments using Mininet for rapid prototyping and testing, and ns-3 for realistic network simulation.

Brain Tumor Detection & Segmentation  **Apr 2026**

- Developed a Deep Learning system to detect brain tumors from MRI images using CNN architectures.
- Implemented a ResNet50 model for tumor classification, achieving **96% accuracy**.
- Built a ResUNet model for segmentation to accurately localize tumor regions using mask prediction.
- For the detection phase, performance is evaluated using Accuracy, Precision, and Recall, while the segmentation phase utilizes Focal Tversky Loss to measure the pixel-level overlap.

Cropify – Crop Yield Prediction  **Jun 2025**

- Prepared a crop yield prediction system using Machine Learning based on environmental and soil features.
- Implemented regression models including Linear Regression, achieving **92% accuracy** with Random Forest.
- Performed data preprocessing and feature engineering to improve model performance.
- Evaluated models using R^2 , MAE and RMSE, selecting Random Forest as the best-performing model.

ACHIEVEMENTS

- Solved 750+ problems on LeetCode in C++, with a strong grasp of Data Structures and Algorithms and a competitive rating of 1500 
- Earned 5-star in Problem Solving, C++, and Python on HackerRank, demonstrating strong algorithmic proficiency 

CERTIFICATIONS

- Machine Learning A-Z: AI, Python [2025] – Udemy (December 2025) 
- Deep Learning Certificate – Infosys Springboard 
- Oracle Certified Foundation Associate – Oracle (October 2025) 

Soft Skills

Problem-Solving, Team Collaboration, Communication, Adaptability, Time Management, Analytical Thinking